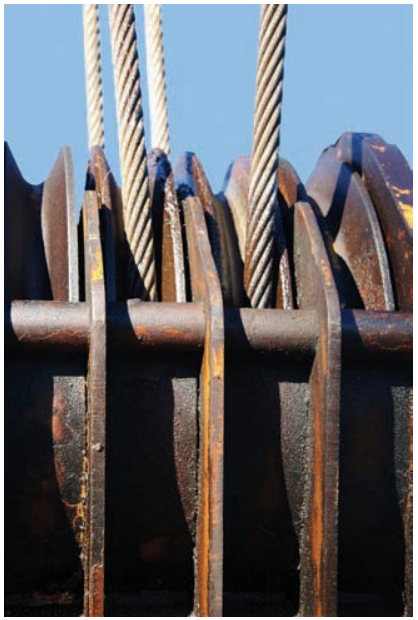
Lecture 2: Axial Loading

Note: content marked in * is more advanced knowledge

Introduction

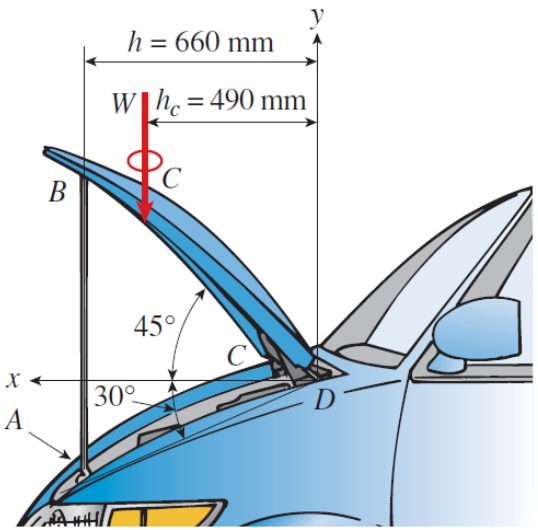




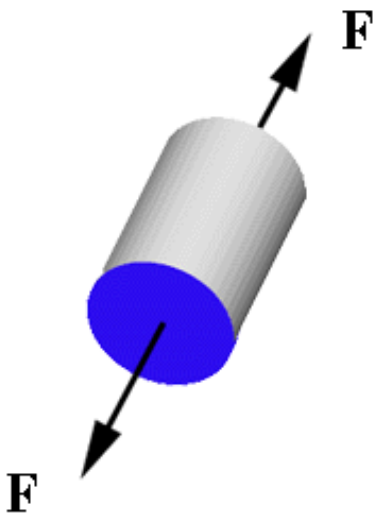


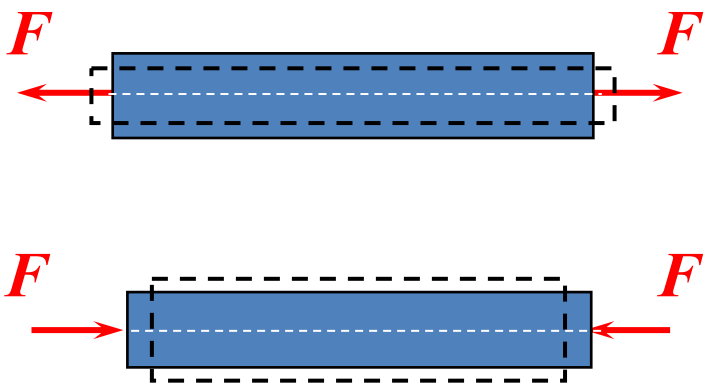






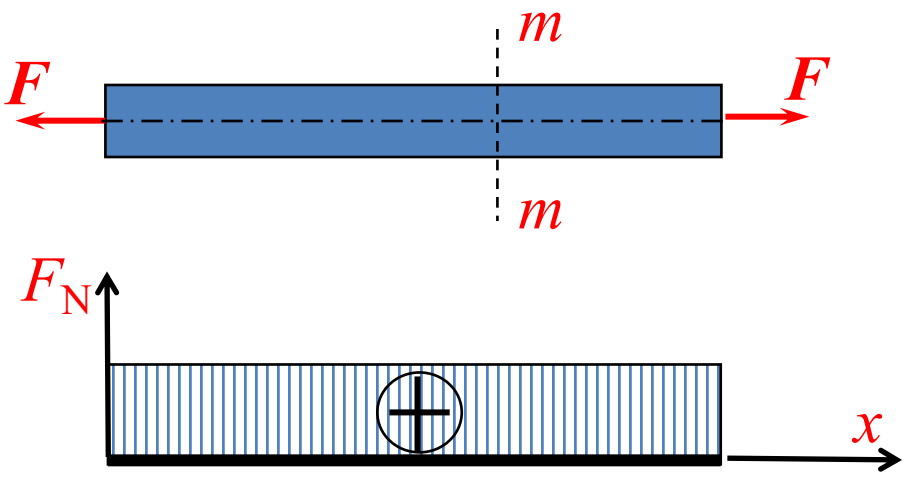
Introduction





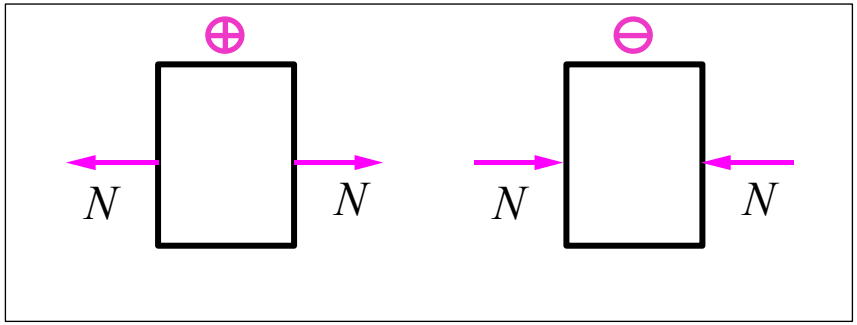
- **Load:** equal and opposite forces along bar axis
- **Deformation:** extension (contraction) along bar axis and contraction (extension) transversely

Diagram of Axial Force



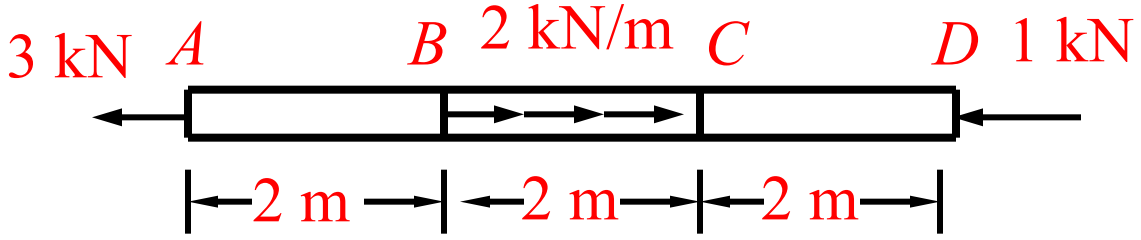
- Positive for tension;

  negative for compression.



Example 1

- Plot the diagram of axial force

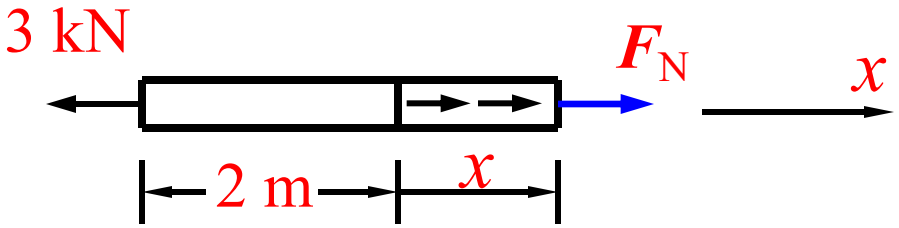


- Solution

1. Internal force in AB、CD

$$F_{\mathrm{N}AB} = 3\ \mathrm{kN}$$

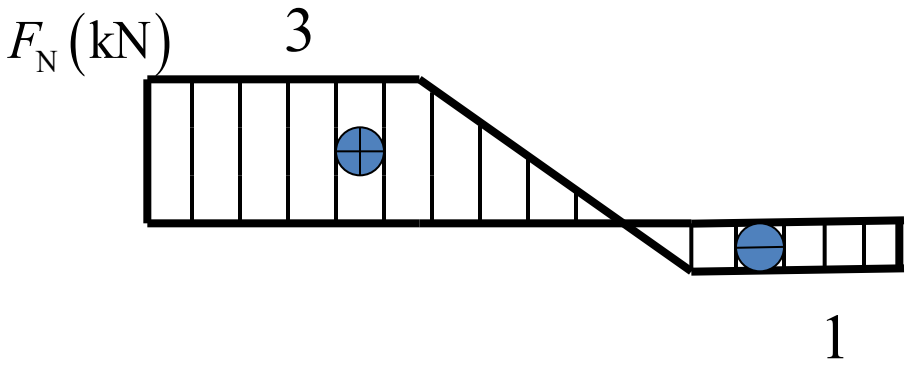
$$F_{\mathrm{N}CD} = -1\ \mathrm{kN}$$



2. Internal force in BC

$$F_{\mathrm{N}}(x) = 3 - 2x \quad (0 < x < 2)$$

3. Diagram of axial force



4. Maximum internal force

$$\left|F_{\mathrm{N,max}}\right| = 3\ \mathrm{kN}$$

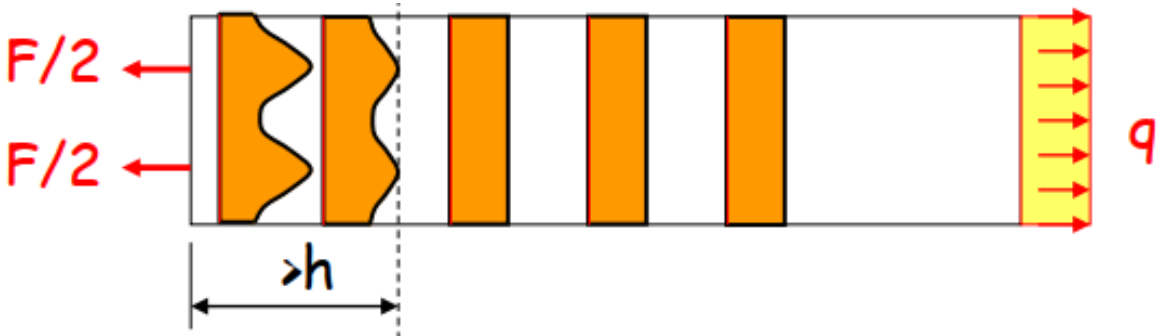
Stresses Acting on Cross Sections*

Saint-Venant's Principle*: For static elastic analysis, the difference between the effects of two different but statically equivalent loads becomes very small at **sufficiently large distances** from the loads.

Therefore, we don't need to study the distribution of external load at the end, as long as it pass the centroid of the beam.

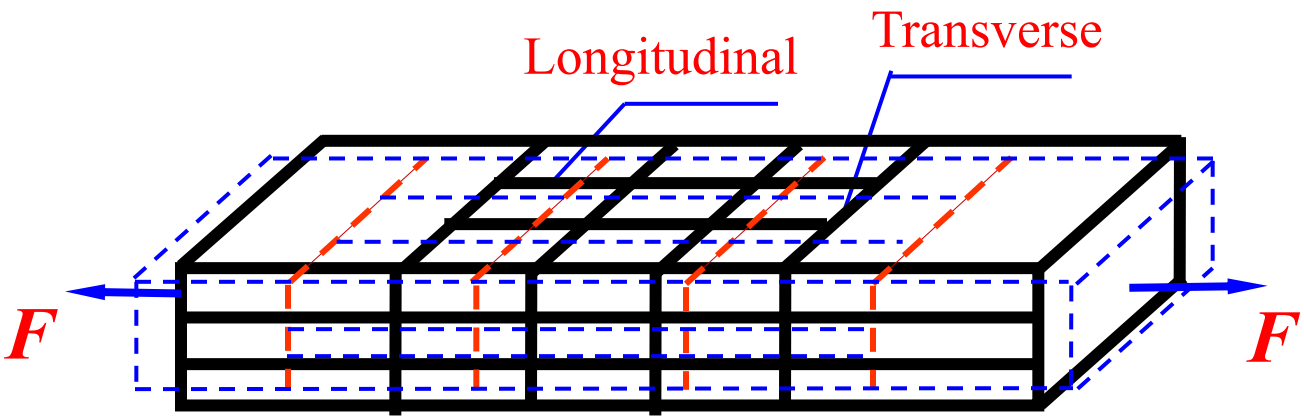


Orange color denotes Normal stress distribution of beam under one point load (from test or more advanced mechanics course)

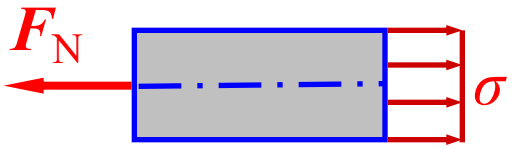


Normal stress distribution of beam under two point loads

Stresses Acting on Cross Sections



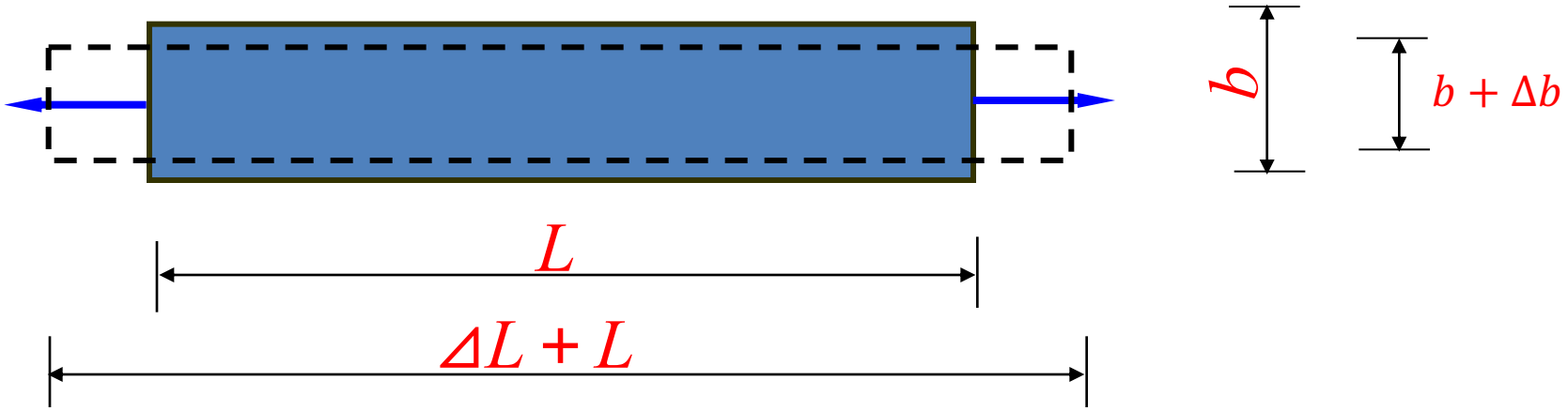
Assumption: stresses are **uniformly distributed** on the cross sections of axially loaded bars (Saint-Venant's Principle)



$$F_N = \int_A \sigma \mathrm{d}A = \sigma \int_A \mathrm{d}A = \sigma A \quad \Rightarrow \quad \sigma = \frac{F_N}{A}$$

Deformation of Axially Loaded Bars

- Longitudinal (axial) & transverse strains



- Longitudinal strain:

$$\varepsilon = \Delta L / L$$

- Transverse strain:

$$\varepsilon_t = \Delta b / b$$

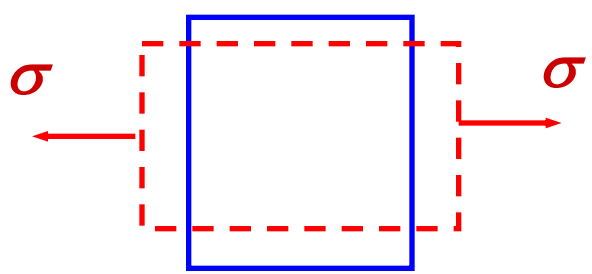
- Relationship between longitudinal and transverse strain

$$\nu = -\varepsilon_t / \varepsilon \qquad \varepsilon_t = -\nu\varepsilon$$

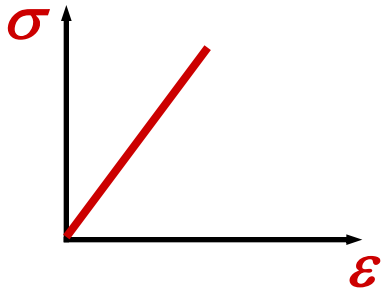
- ν: Poisson's ratio: 0.2 for concrete, 0.3 for steel before yielding, close to 0.5 for incompressible material (rubber).

Deformation of Axially Loaded Bars

- For uniaxial loading,



$$\sigma \equiv E\varepsilon$$



- This linear relation is referred to as the one-dimensional **Hooke's law**, with E denoting the Young's modulus.

- The measurement of axial deformation

$$\Delta L = L\varepsilon = L\frac{\sigma}{E} = \frac{LF_{\mathrm{N}}}{AE} = \frac{F_{\mathrm{N}}L}{EA}$$

- EA: section tension (Compression) stiffness, A: Cross section area

Elastic Constants of Engineering Materials

Material	Modulus of elasticity E		Shear modulus of elasticity G		Poisson's ratio ν
	ksi	GPa	ksi	GPa	
Aluminum alloys	10,000–11,400	70–79	3,800–4,300	26–30	0.33
2014-T6	10,600	73	4,000	28	0.33
6061-T6	10,000	70	3,800	26	0.33
7075-T6	10,400	72	3,900	27	0.33
Brass	14,000–16,000	96–110	5,200–6,000	36–41	0.34
Bronze	14,000–17,000	96–120	5,200–6,300	36–44	0.34
Cast iron	12,000–25,000	83–170	4,600–10,000	32–69	0.2–0.3
Concrete (compression)	2,500–4,500	17–31			0.1–0.2
Copper and copper alloys	16,000–18,000	110–120	5,800–6,800	40–47	0.33–0.36
Glass	7,000–12,000	48–83	2,700–5,100	19–35	0.17–0.27
Magnesium alloys	6,000–6,500	41–45	2,200–2,400	15–17	0.35
Monel (67% Ni, 30% Cu)	25,000	170	9,500	66	0.32

No need to remember the numbers of this table.

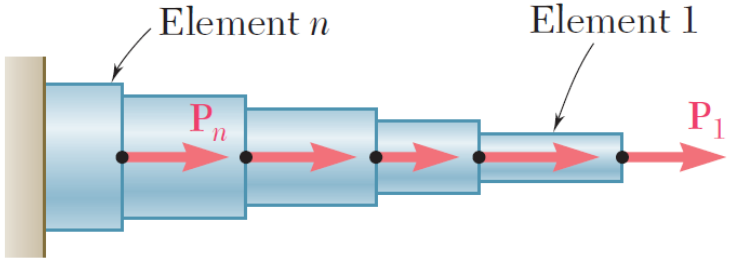
Elastic Constants of Engineering Materials

Material	Modulus of elasticity E		Shear modulus of elasticity G		Poisson's ratio ν
	ksi	GPa	ksi	GPa	
Nickel	30,000	210	11,400	80	0.31
Plastics					
Nylon	300–500	2.1–3.4			0.4
Polyethylene	100–200	0.7–1.4			0.4
Rock (compression)					
Granite, marble, quartz	6,000–14,000	40–100			0.2–0.3
Limestone, sandstone	3,000–10,000	20–70			0.2–0.3
Rubber	0.1–0.6	0.0007–0.004	0.03–0.2	0.0002–0.001	0.45–0.50
Steel	28,000–30,000	190–210	10,800–11,800	75–80	0.27–0.30
Titanium alloys	15,000–17,000	100–120	5,600–6,400	39–44	0.33
Tungsten	50,000–55,000	340–380	21,000–23,000	140–160	0.2
Wood (bending)					
Douglas fir	1,600–1,900	11–13			
Oak	1,600–1,800	11–12			
Southern pine	1,600–2,000	11–14			

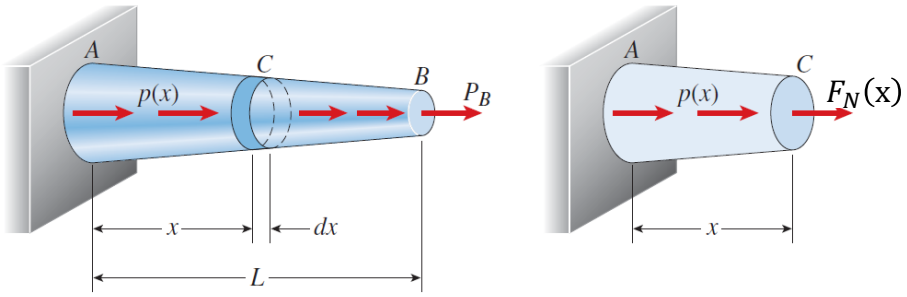
No need to remember the numbers of this table.

Non-uniform Tension/Compression

- Pure tension/compression formula refers to a prismatic bar subjected to axial forces acting only at the ends.
- Nonuniform tension/compression differs from pure tension in that the bar need not to be prismatic and the applied axial forces may act anywhere along the axis of the bar.
- Bars in nonuniform tension/compression can be analyzed by applying the formulas of pure tension/compression to finite segments of the bar and then adding the results, or by applying the formulas to differential elements of the bar and then integrating.



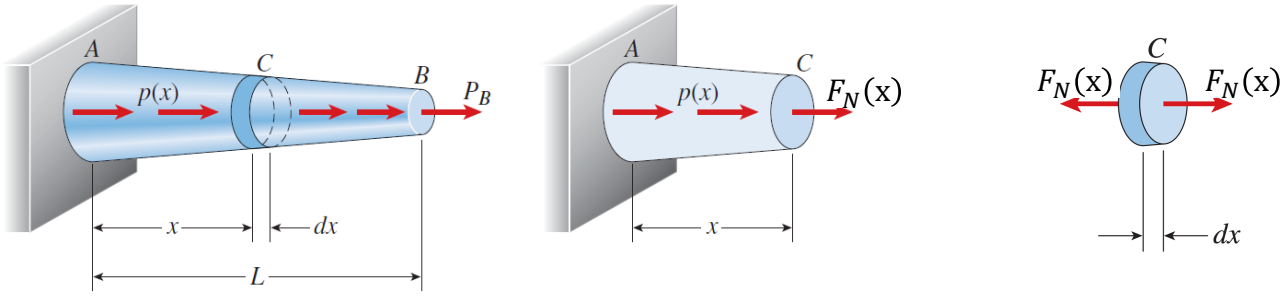
$$\Delta L = \sum_i \Delta L_i = \sum_i \frac{F_{Ni} L_i}{E_i A_i}$$



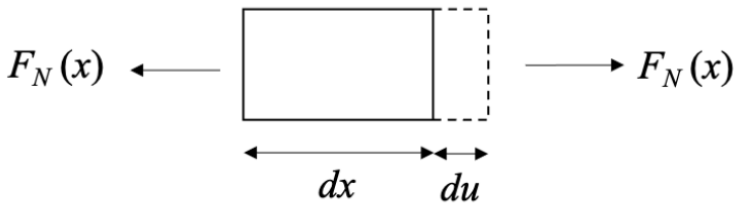
$$\Delta L = \int_0^L d\Delta L = \int_0^L \frac{F_N(x)\,dx}{EA(x)}$$

Governing equation of axial force in beam*

Suppose the area of bar change as $A_0(x)$, a distributed force $p(x)$ is applied with P_B



Axial displacement of differential segment $du(x)$



$$\varepsilon(x) = \frac{\sigma(x)}{E}$$

$$\frac{du(x)}{dx} = \frac{F_N(x)}{EA_0(x)}$$

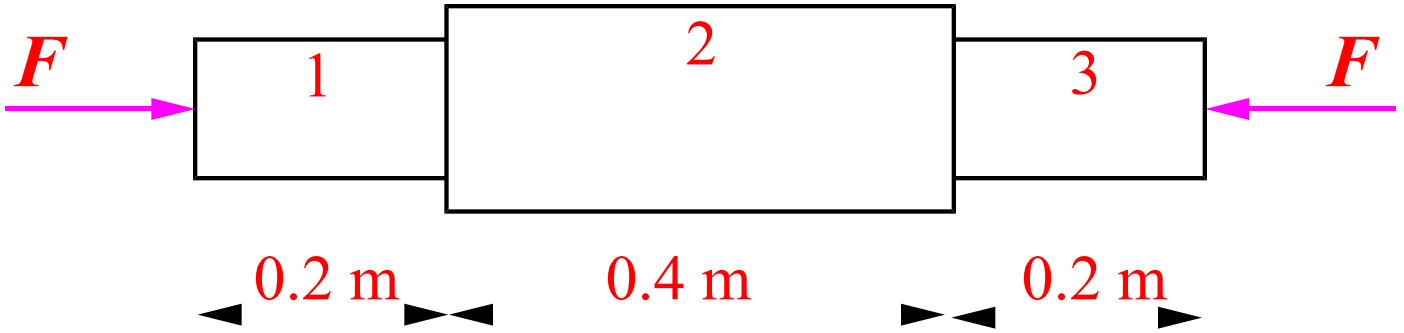
$$du(x) = \frac{F_N(x)}{EA_0(x)}dx$$

$$\Delta L = \int_0^L du(x) = \int_0^L \frac{F_N(x)}{EA_0(x)}dx$$

Note:
$F_N(x)$: internal axial force
$u(x)$: axial displacement
$p(x)$: axial distributed load per unit length
P_B: external force applied at end B
L: total length of member

Example 2

- Consider an axially loaded bar with varied cross section. Given: E = 210 GPa; Section 1 (circular): d_1 = 20 mm; Section 2 (square): side length a = 25 mm, σ_2 = -30 MPa; Section 3 (circular) d_3=12mm. Find: Total change in bar length $\triangle L$.



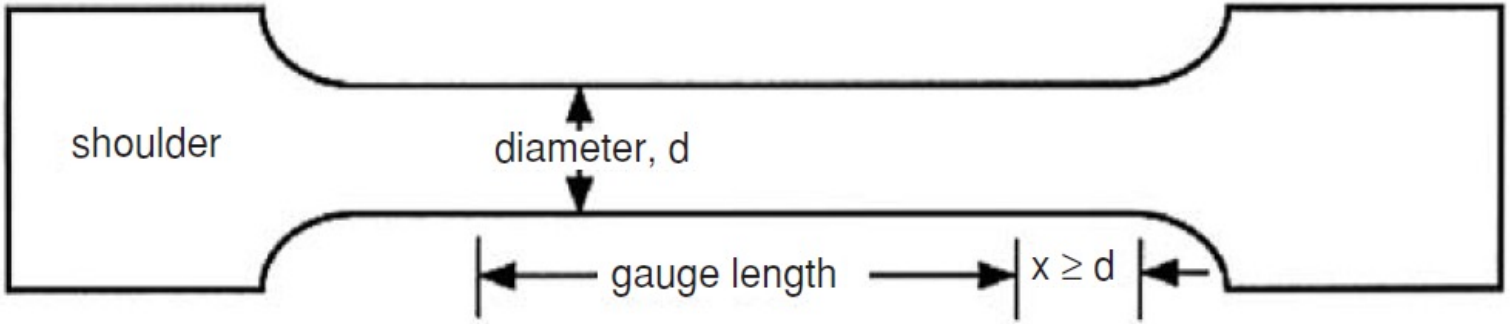
- Solution:

$$F = \sigma_2 A_2 = -30\ \text{MPa} \times 25^2\ \text{mm}^2 = -18.75\ \text{kN}$$

$$\Delta L = \frac{F_{N_1} L_1}{EA_1} + \frac{F_{N_2} L_2}{EA_2} + \frac{F_{N_3} L_3}{EA_3} = \frac{-18750}{210\times10^9}\left(\frac{0.2}{\frac{\pi\times0.02^2}{4}} + \frac{0.4}{0.025^2} + \frac{0.2}{\frac{\pi\times0.012^2}{4}}\right) = -0.272\text{mm}$$

Mechanical Behavior of Materials

- Mechanical behavior of materials focus on the strength and deformation characteristics of solid materials under external loading.
- Tensile specimen



Typical tensile specimen with a reduced gauge section and larger shoulders.

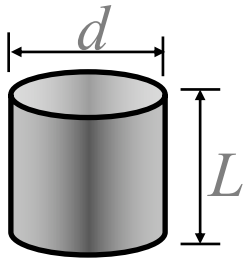
- For cylindrical tensile specimen

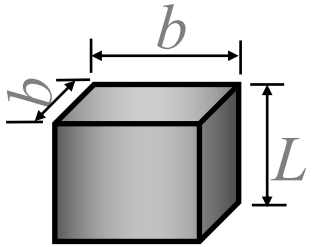
  $L = 10d$ *or* $5d$

- For square tensile specimen

  $L = 11.3b$ or $5.65b$.

- Compressive specimen: $L = 1\sim3d,\ 1\sim3b$

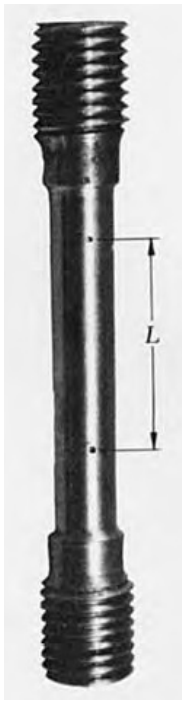




Mechanical Behavior of Materials



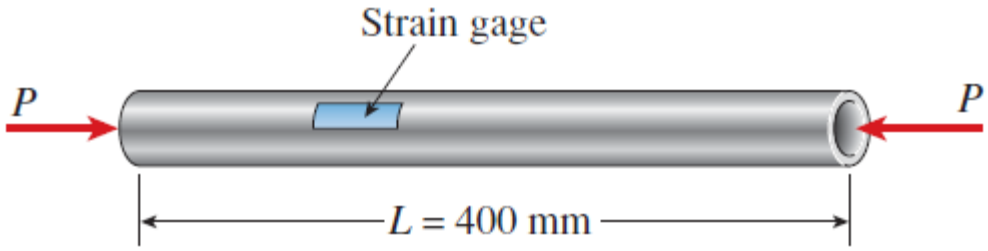




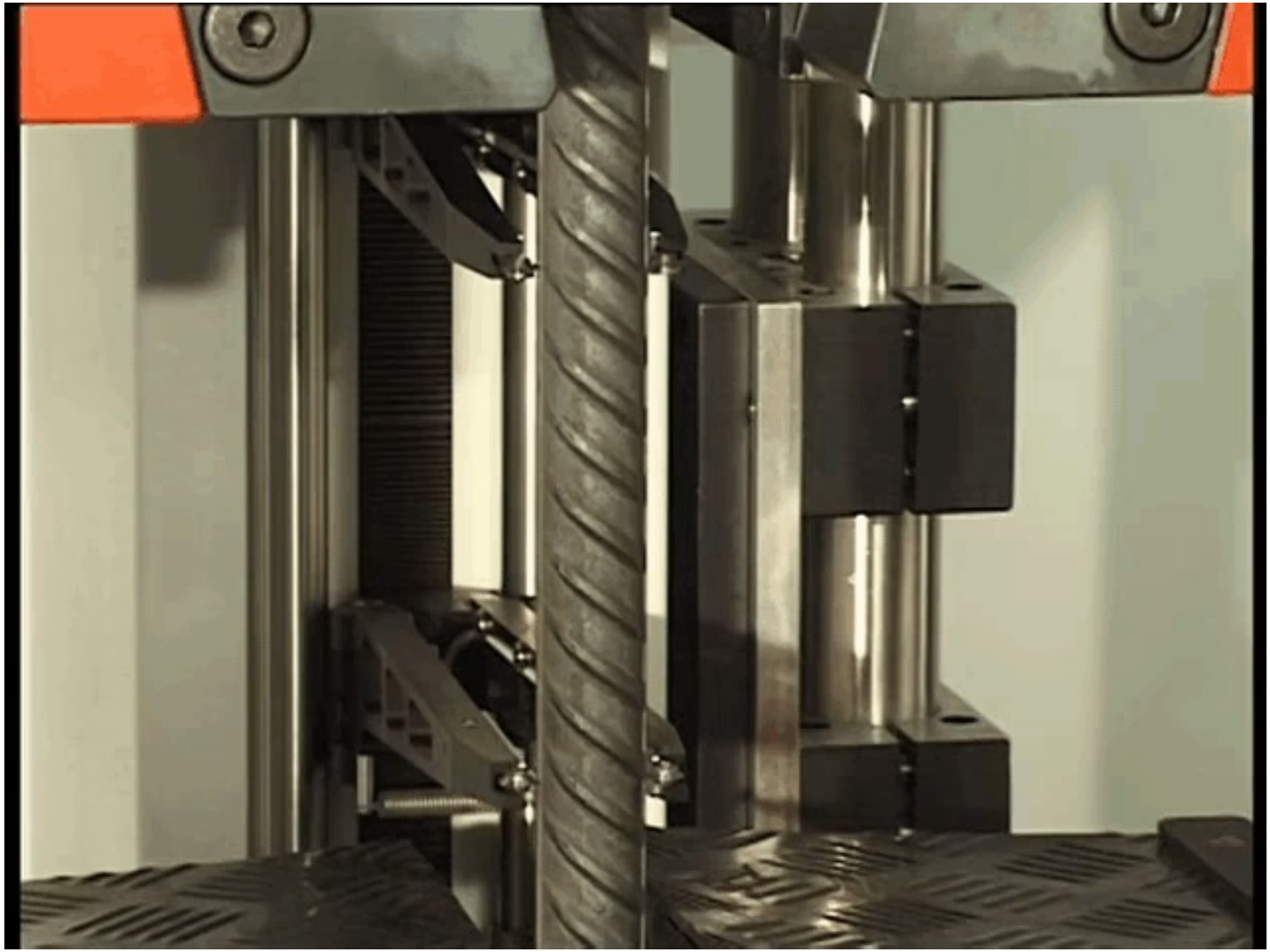






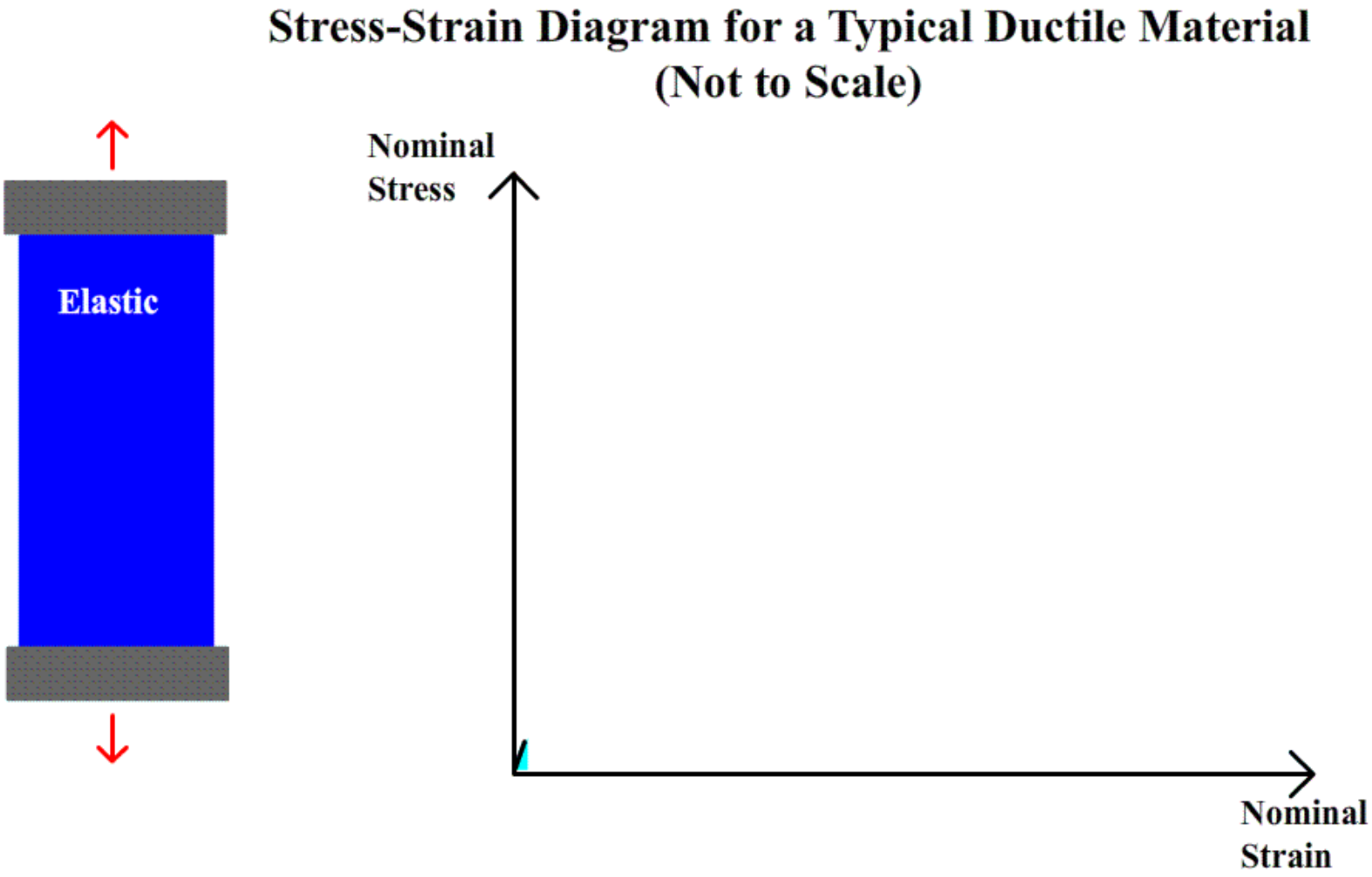


Mechanical Behavior of Materials



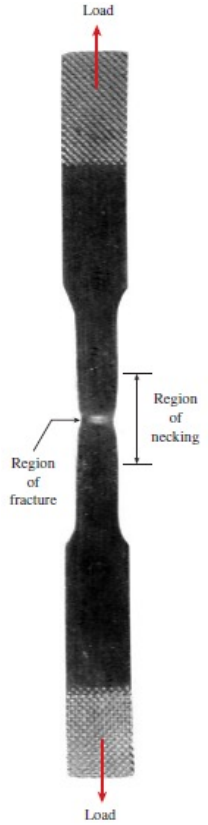
- Uniaxial tension test of reinforcing bar

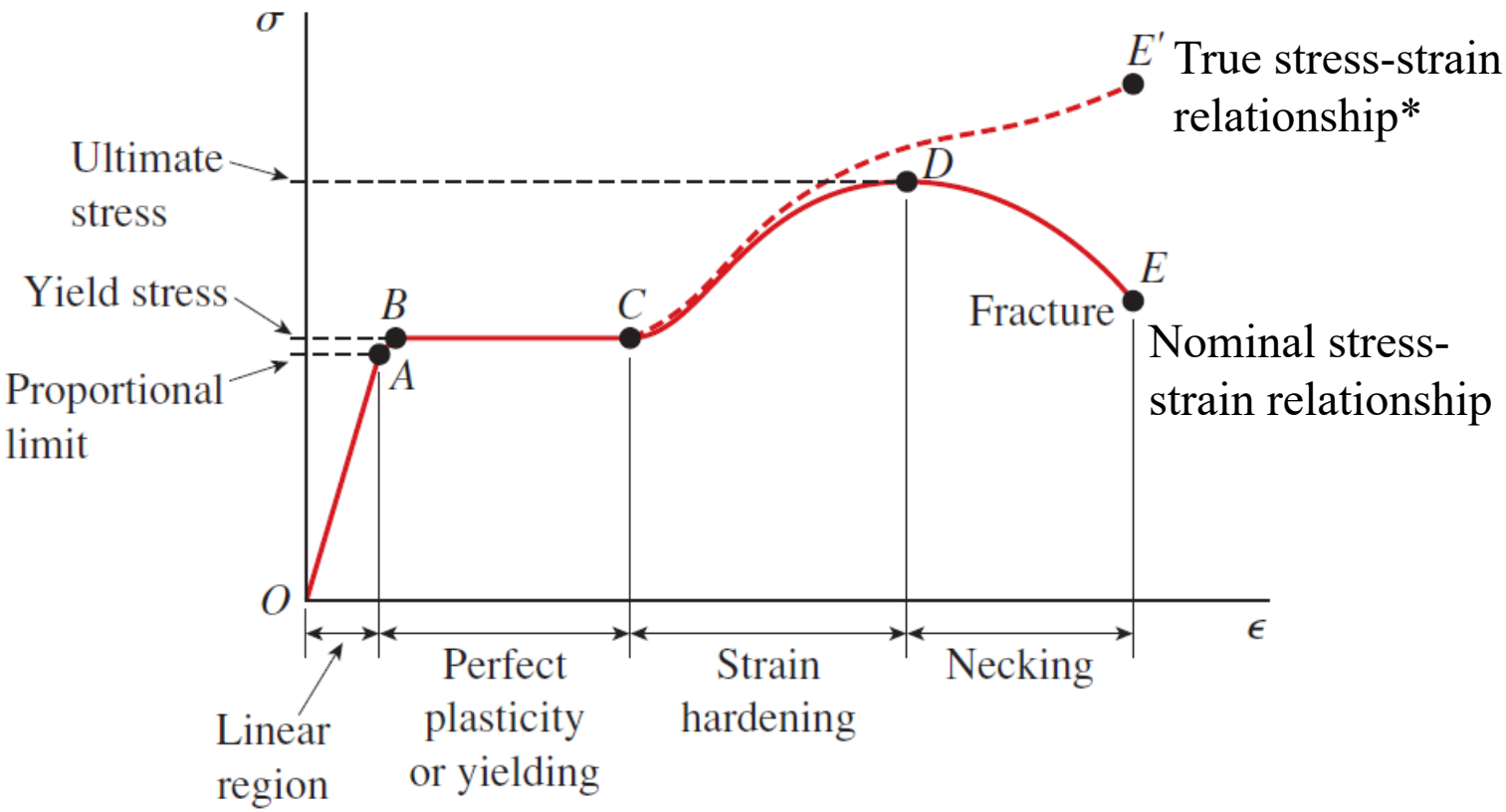
Nominal Stress-strain Curve



- Find more animations at http://em2lab.yolasite.com/.

Stress and Deformation Indices of Low-carbon Steel





- σ_p: Proportional limit
- σ_Y: Yield stress (yield strength)
- σ_u: Ultimate stress (ultimate strength)

- Percent elongation:

$$\delta = (L_{\text{final}} - L_{\text{initial}}) / L_{\text{initial}} \times 100\%$$

- Percent reduction in area

$$\psi = (A_{\text{initial}} - A_{\text{final}}) / A_{\text{initial}} \times 100\%$$

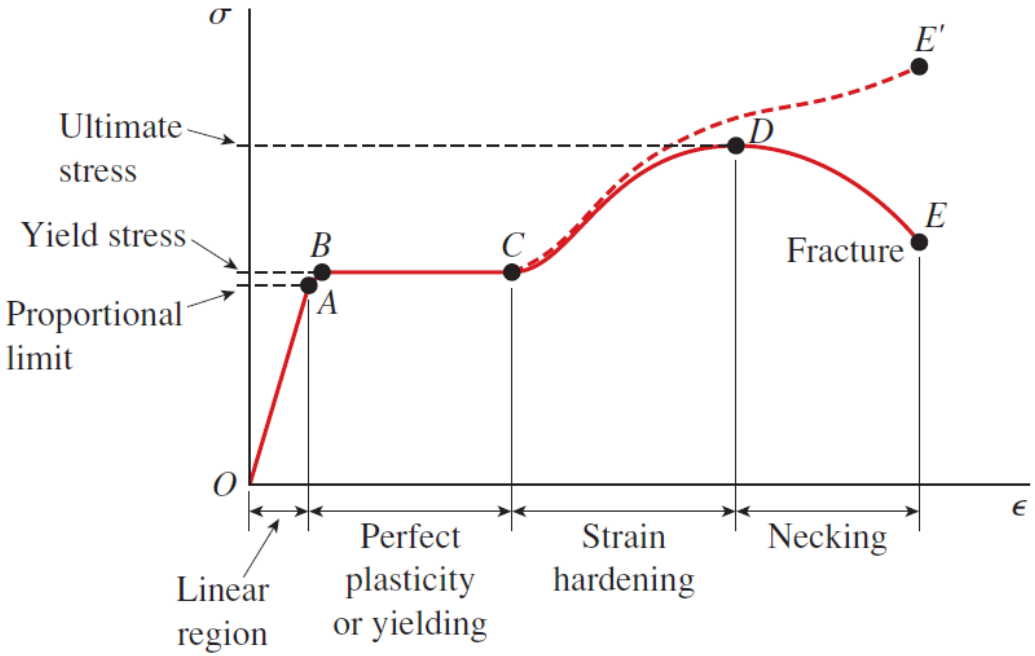
True stress vs nominal stress*

- True stress (Cauchy stress): axial force F divided by **current** cross-section area A **after deformation**

$$\sigma_{\text{true}} = \frac{F}{A}$$

- Engineering stress (nominal): axial force F divided by **original** cross-section area A_0

$$\sigma = \frac{F}{A_0}$$



Yield Stress, Ultimate Stress and Percent Elongation

Material	Yield stress σ_Y		Ultimate stress σ_U		Percent elongation (2 in. gage length)
	ksi	MPa	ksi	MPa	
Aluminum alloys 2014-T6 6061-T6 7075-T6	5–70 60 40 70	35–500 410 270 480	15–80 70 45 80	100–550 480 310 550	1–45 13 17 11
Brass	10–80	70–550	30–90	200–620	4–60
Bronze	12–100	82–690	30–120	200–830	5–60
Cast iron (tension)	17–42	120–290	10–70	69–480	0–1
Cast iron (compression)			50–200	340–1,400	
Concrete (compression)			1.5–10	10–70	
Copper and copper alloys	8–110	55–760	33–120	230–830	4–50
Glass Plate glass Glass fibers			5–150 10 1,000–3,000	30–1,000 70 7,000–20,000	0
Magnesium alloys	12–40	80–280	20–50	140–340	2–20
Monel (67% Ni, 30% Cu)	25–160	170–1,100	65–170	450–1,200	2–50
Nickel	15–90	100–620	45–110	310–760	2–50

No need to remember the numbers of this table.

Yield Stress, Ultimate Stress and Percent Elongation

Material	Yield stress σ_Y		Ultimate stress σ_U		Percent elongation (2 in. gage length)
	ksi	MPa	ksi	MPa	
Steel wire	40–150	280–1,000	80–200	550–1,400	5–40
Titanium alloys	110–150	760–1,000	130–170	900–1,200	10
Tungsten			200–600	1,400–4,000	0–4
Wood (bending)					
Douglas fir	5–8	30–50	8–12	50–80	
Oak	6–9	40–60	8–14	50–100	
Southern pine	6–9	40–60	8–14	50–100	
Wood (compression parallel to grain)					
Douglas fir	4–8	30–50	6–10	40–70	
Oak	4–6	30–40	5–8	30–50	
Southern pine	4–8	30–50	6–10	40–70	

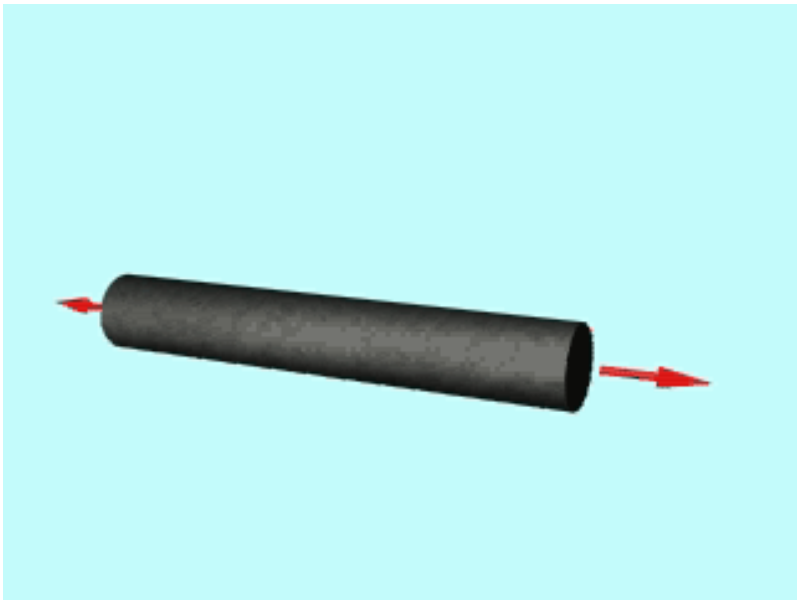
No need to remember the numbers of this table.

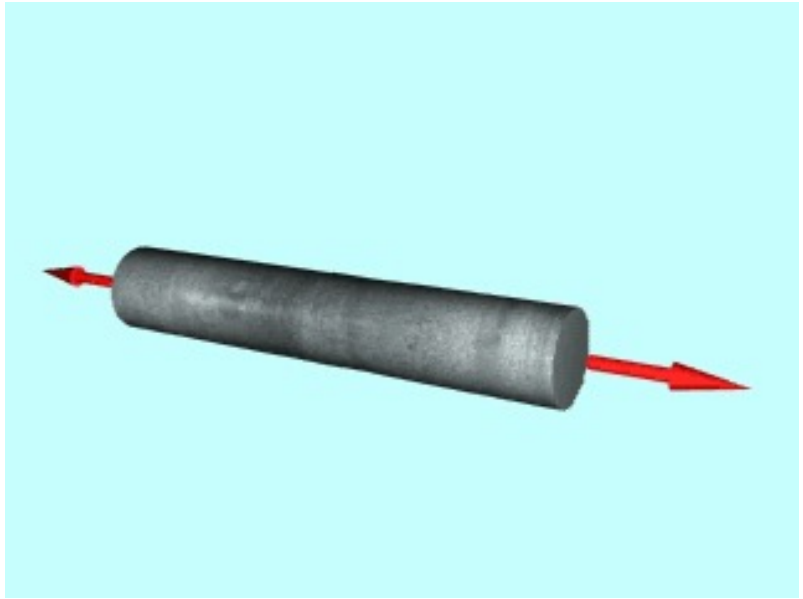
Yield Stress, Ultimate Stress and Percent Elongation

Material	Yield stress σ_Y		Ultimate stress σ_U		Percent elongation (2 in. gage length)
	ksi	MPa	ksi	MPa	
Plastics					
Nylon			6–12	40–80	20–100
Polyethylene			1–4	7–28	15–300
Rock (compression)					
Granite, marble, quartz			8–40	50–280	
Limestone, sandstone			3–30	20–200	
Rubber	0.2–1.0	1–7	1–3	7–20	100–800
Steel					
High-strength	50–150	340–1,000	80–180	550–1,200	5–25
Machine	50–100	340–700	80–125	550–860	5–25
Spring	60–240	400–1,600	100–270	700–1,900	3–15
Stainless	40–100	280–700	60–150	400–1,000	5–40
Tool	75	520	130	900	8
Steel, structural	30–100	200–700	50–120	340–830	10–40
ASTM-A36	36	250	60	400	30
ASTM-A572	50	340	70	500	20
ASTM-A514	100	700	120	830	15

No need to remember the numbers of this table.

Failure of Brittle vs. Ductile Bars under Tension



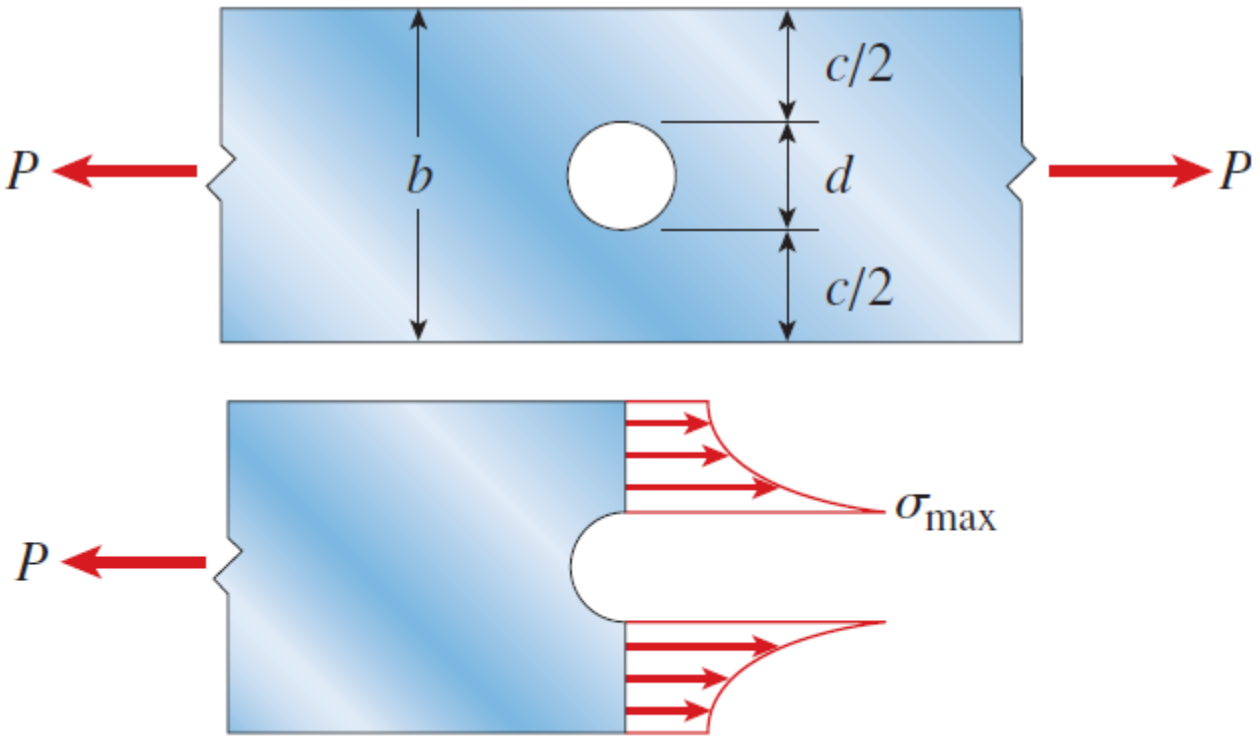


Brittle

Ductile

- Under uniaxial tension, brittle bars break along cross-sections while ductile ones glide along 45^0 sections during yielding.

Stress Concentration*



- **Stress concentration:** rapid stress increase at specific locations where geometric defects and/or abrupt cross-sectional area change occurs

Strength Condition (Allowable stress design, ASD)

- **Limit stress ($\sigma_{\lim}$):** the stress under which mechanical components are damaged.
- **Damage Criteria:** yielding (σ_Y) for ductile materials; fracture (σ_u) for brittle materials
- **Allowable Stress [σ]:** the maximum stress allowed in engineering practice. It is typically taken as one nth (Safety Factor) of the limit stress
- **Strength Condition:**

$$\sigma_{\max} \le \frac{\sigma_{\lim}}{n} = [\sigma]$$

- **Strength Analysis:**

(1) Strength check: $\sigma_{\max} \le [\sigma]$

(2) Cross-section design: $A \ge F_{N\max} / [\sigma]$

(3) Find allowable load: $F_{N\max} \le A[\sigma]$

Strength Condition (Example 3)

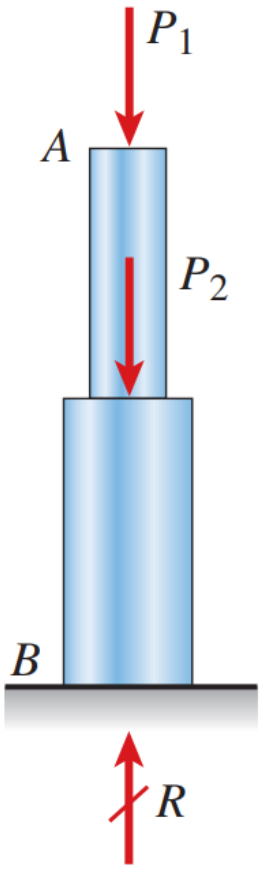
Given: A circular bar has diameter d = 14 mm, allowable strength [σ] = 17.0 MPa, the uniaxial tension load F = 2.5 kN is applied to the bar. Check the strength condition of the circular bar.

- Solution

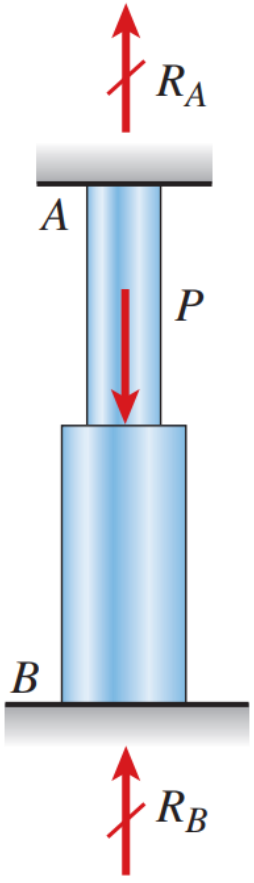
$$\sigma_{max} = \frac{F_{N,max}}{A} = \frac{2.5\times10^{3}\ N}{\frac{\pi}{4}\times14^{2}\times10^{-6}m^{2}} = 16.2\ \text{MPa} \leq [\sigma]$$

- The strength condition is satisfied.

Statically Indeterminate Structure



Statically determinate



Statically indeterminate

Statically indeterminate bars

To solve for statically indeterminate bars, there is an additional unknown (reaction force at A), **equation of force equilibrium** will not solve it alone, additional equation is needed to solve it:

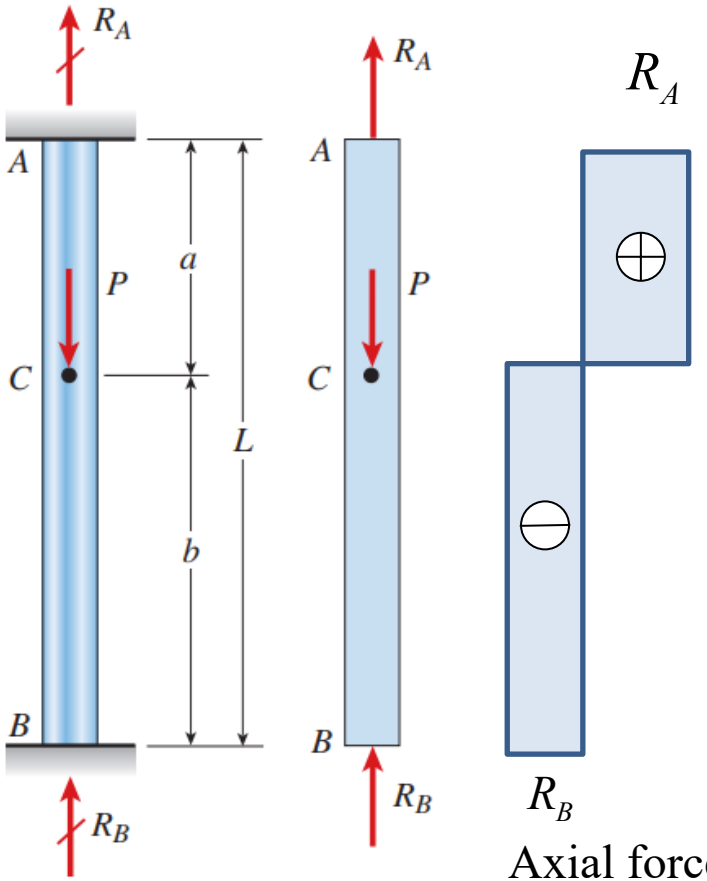
$$\Sigma F_{\text{vert}} = 0 \quad R_A - P + R_B = 0$$

The additional equation is the total length of the bar is not changed during the loading (**equation of compatibility**):

$$\Delta L = 0$$

Force-displacement relationship: calculate the displacement of bar based on internal force

Statically Indeterminate Structure



Axial force diagram

To solve the statically indeterminate bars, the section stiffness need to be input, assume the bar has uniform EA and total length L, a point load P is applied at point C.

1. Force equilibrium

$$\Sigma F_{\text{vert}} = 0 \quad R_A - P + R_B = 0$$

2. Force-displacement relationship

$$\Delta_{AC} = \frac{R_A a}{EA} \qquad \Delta_{CB} = -\frac{R_B b}{EA}$$

3. Equation of compatibility

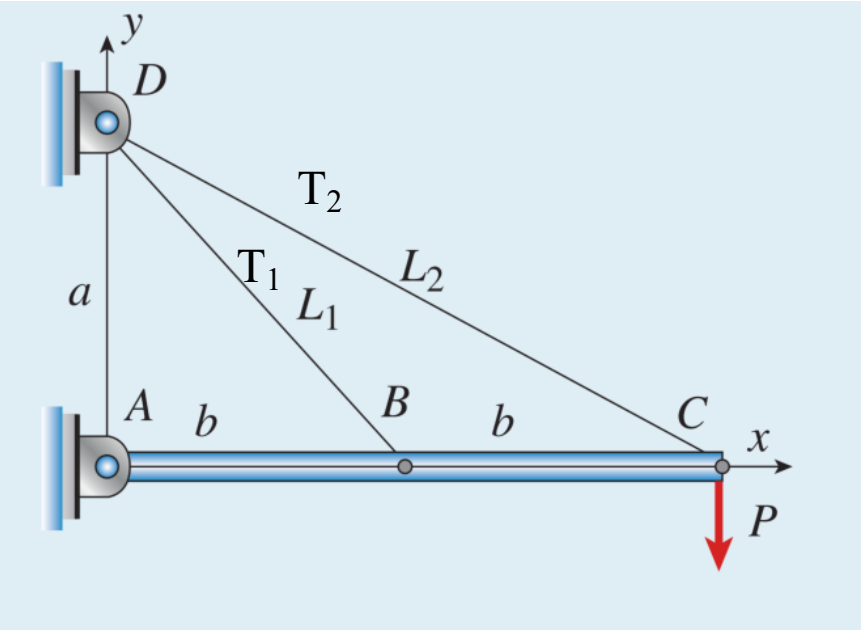
$$\Delta L = \Delta_{AC} + \Delta_{CB} = \frac{R_A a}{EA} - \frac{R_B b}{EA} = 0$$

$$R_A = \frac{Pb}{L} \qquad R_B = \frac{Pa}{L}$$

$$\delta_C = \Delta_{AC} = \frac{R_A a}{EA} = \frac{Pab}{LEA}$$

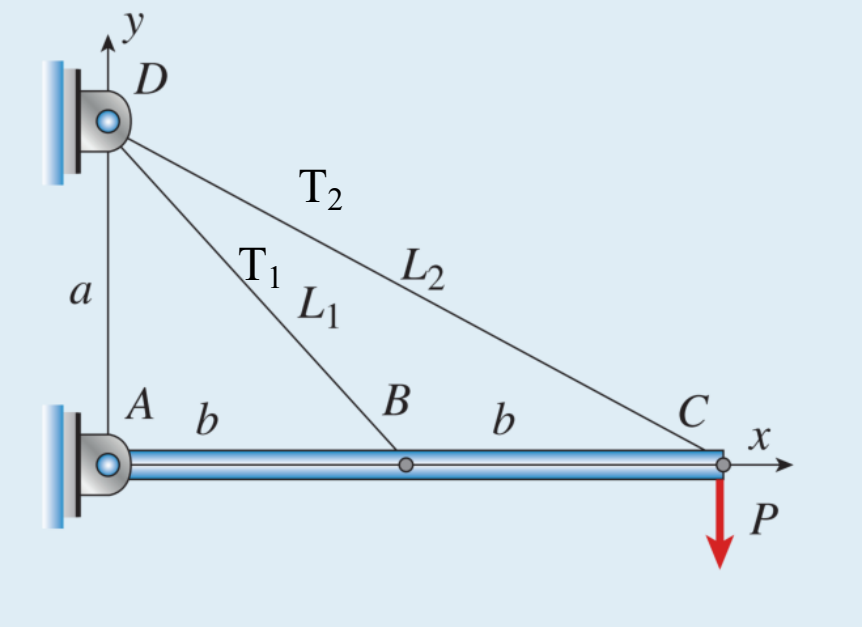
Statically Indeterminate Structure (Example 4)

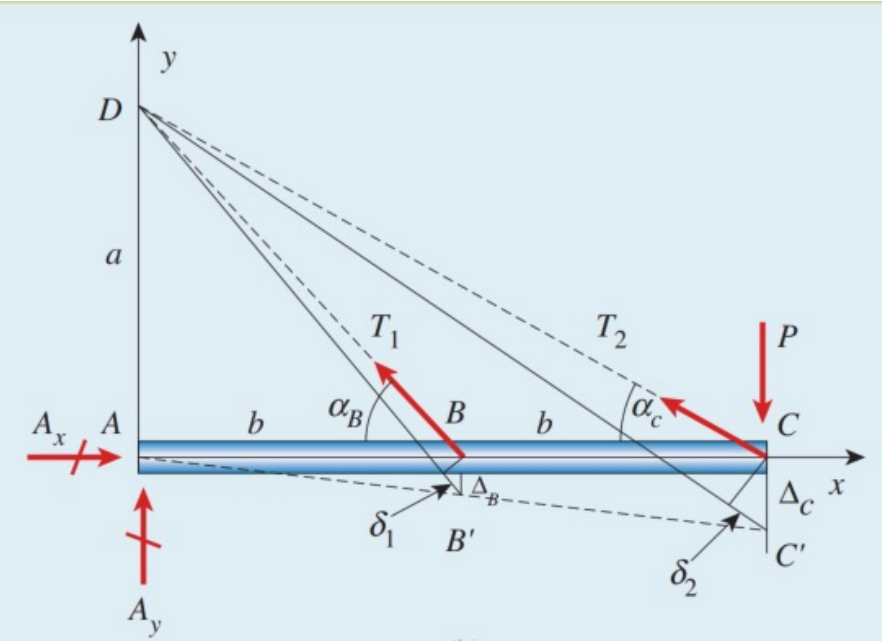
Question: A horizontal **rigid** bar ABC is pinned at A and supported by two wires DB and DC at points B and C. A vertical load P act on point C at bar end. All wires have modulus of E and area of A. Obtain the expression for the internal force of the wires T_1 and T_2.



Note: In rigid bar ABC, axial force, shear force and moment can happen, but we neglect the displacement induced by these force and moment and only consider rigid body rotation around hinge A in this study.

Statically Indeterminate Structure (Example 4)



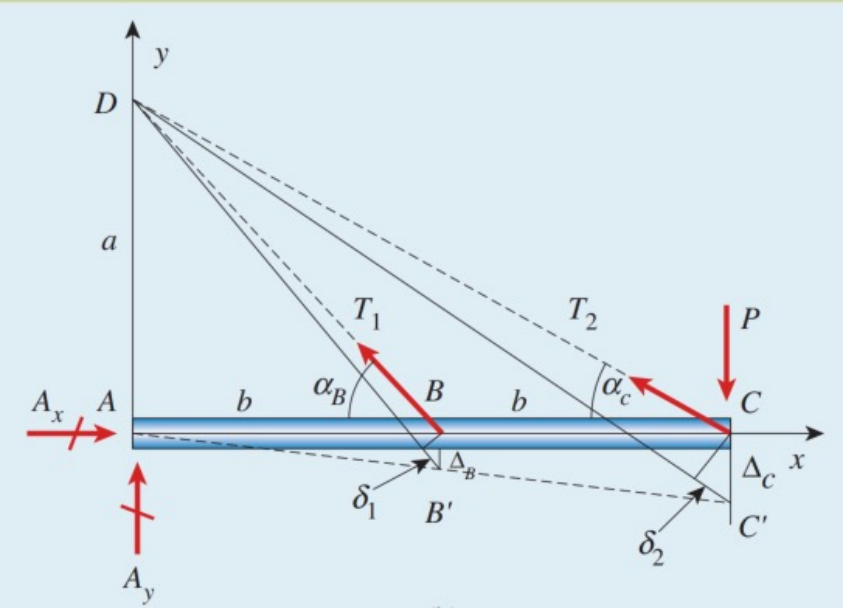


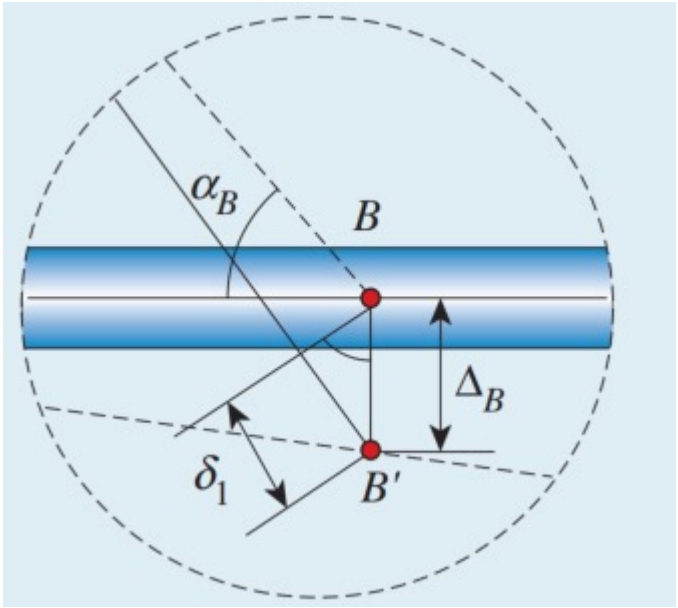
1. Equation of equilibrium: Taking moments about point A (with counterclockwise moments being positive)

$$\Sigma M_A = 0 \quad T_1 \cdot b \cdot \sin\left(\alpha_B\right) + T_2 \cdot 2b \cdot \sin\left(\alpha_C\right) - P \cdot 2b = 0$$

$$T_1 \cdot \sin\left(\alpha_B\right) + T_2 \cdot 2\sin\left(\alpha_C\right) = 2P$$

Statically Indeterminate Structure (Example 4)





2. Compatibility equation: Because ABC is rigid bar, it can only rotate around A. The deflection at C must be twice the deflection at B.

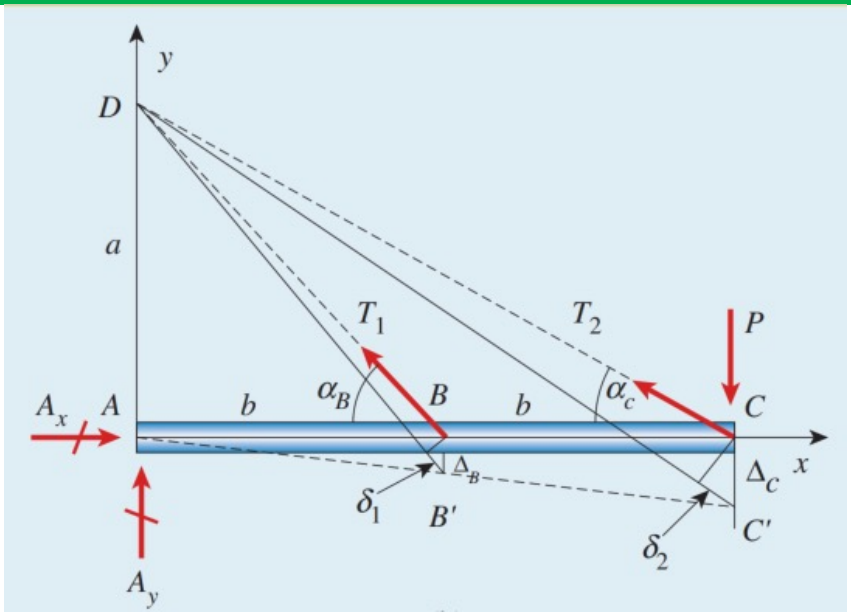
$$\Delta_C = 2\Delta_B$$

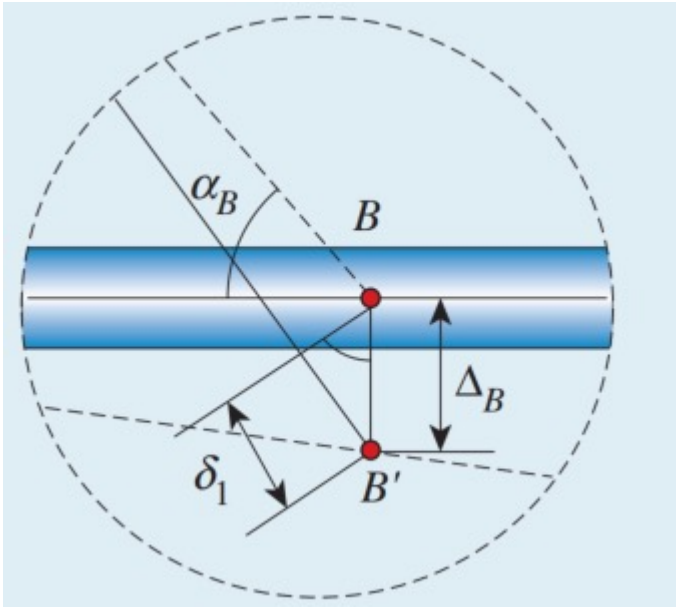
Using geometry relationship at point B, the elongation of cable DB can be obtained as

$$\delta_1 = \Delta_B \sin(\alpha_B) \qquad \delta_2 = \Delta_C \sin(\alpha_C) = 2\Delta_B \sin(\alpha_C)$$

$$\delta_2 = 2\delta_1 \frac{\sin(\alpha_C)}{\sin(\alpha_B)}$$

Statically Indeterminate Structure (Example 4)





3. Force-displacement relationship (Hooke's law):

$$\delta_1 = \left(\frac{L_1}{EA}\right)T_1 \qquad \delta_2 = \left(\frac{L_2}{EA}\right)T_2 \qquad \delta_2 = 2\delta_1 \frac{\sin(\alpha_C)}{\sin(\alpha_B)}$$

$$T_2 = 2\left(\frac{\sin(\alpha_C)}{\sin(\alpha_B)}\right)\left(\frac{L_1}{L_2}\right)T_1 = 2\left(\frac{\sin(\alpha_C)}{\sin(\alpha_B)}\right)\left(\frac{a/\sin(\alpha_B)}{a/\sin(\alpha_C)}\right)T_1 = 2\left(\frac{\sin(\alpha_C)}{\sin(\alpha_B)}\right)^2 T_1$$

Recall $\quad T_1 \cdot \sin(\alpha_B) + T_2 \cdot 2\sin(\alpha_C) = 2P$

$$\Rightarrow \quad T_1 = \frac{2P\sin(\alpha_B)^2}{\sin(\alpha_B)^3 + 4\sin(\alpha_C)^3} \qquad T_2 = \frac{4P\sin(\alpha_C)^2}{\sin(\alpha_B)^3 + 4\sin(\alpha_C)^3}$$

Statically Indeterminate Structure (Example 5)

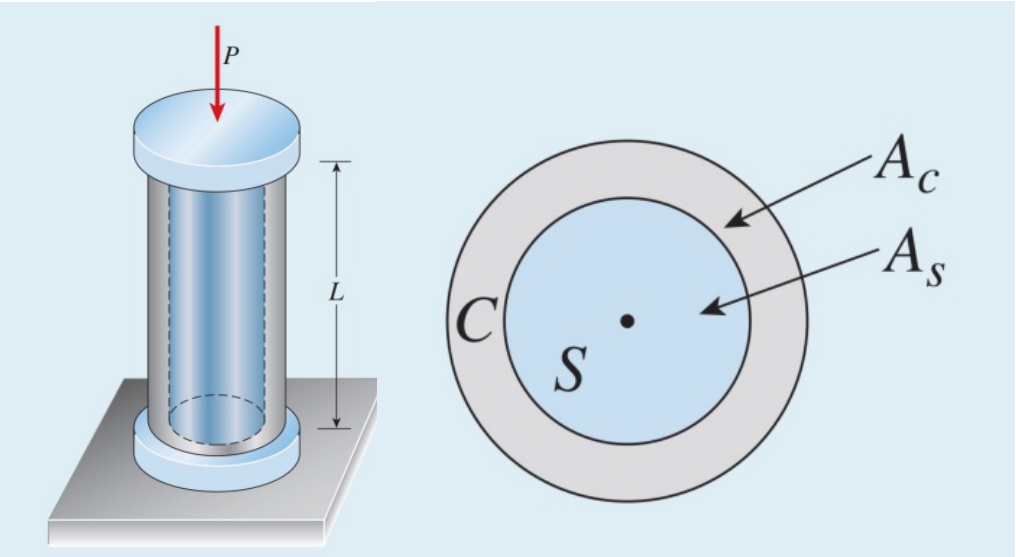
Question: A solid circular steel cylinder S is encased in a hollow concrete tube C. The cylinder and tube are compressed between the rigid plates of a testing machine by compressive forces P. The steel cylinder has cross-sectional area A_s and modulus of elasticity E_s, the concrete tube has area A_c and modulus E_c, and both parts have length L. Assume the Poisson ratio of these two material are the same (although they are not same in reality).

Determine:

(a) compressive forces P_s in steel and P_c in concrete.

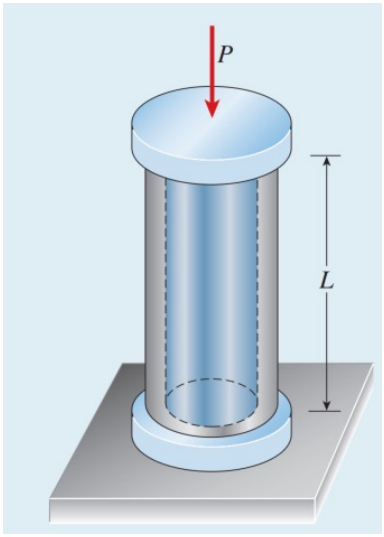
(b) Compressive stress in steel and concrete;

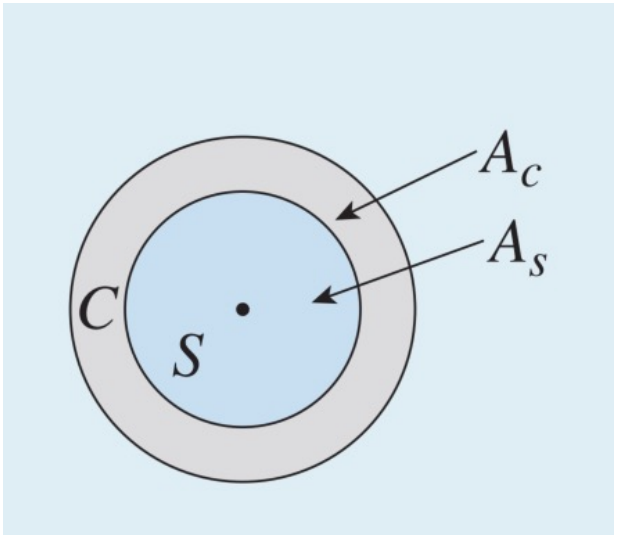
(c) shortening of member

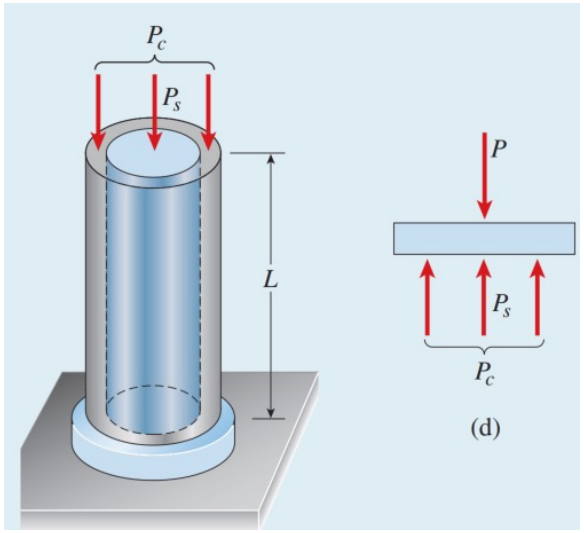


Composite column

Statically Indeterminate Structure (Example 5)







Note: because we assume the Poisson ratio of inner and outer material are the same, both material will be under uniaxial compression. Otherwise, the transverse direction has additional constraint and special consideration need to be included to consider the lateral strain (transverse strain of both material).

Force Equilibrium equation: $P_S + P_C = P$

Strain compatibility equation: the shortening of the steel and concrete parts are the same

$$\delta_s = \delta_c$$

Force-displacement relationship: $\delta_s = \dfrac{P_s L}{E_s A_s} \quad \delta_c = \dfrac{P_c L}{E_c A_c}$

Statically Indeterminate Structure (Example 5)

Solving the equations obtain:

$$P_s = P\left(\frac{E_s A_s}{E_s A_s + E_c A_c}\right) \quad P_c = P\left(\frac{E_c A_c}{E_s A_s + E_c A_c}\right)$$

From force-displacement relationship, the longitudinal deformation can be obtained:

$$\delta_s = \delta_c = \frac{PL}{E_s A_s + E_c A_c}$$

Stress in components: $\sigma_s = \frac{P_s}{A_s} = \frac{PE_s}{E_s A_s + E_c A_c}$ $\sigma_c = \frac{P_c}{A_c} = \frac{PE_c}{E_s A_s + E_c A_c}$

- Note: the ratio σ_s/σ_c is equal to the ratio E_s/E_c of the modulus of elasticity, meaning that, in general, the "stiffer" materials always has the larger stress.

Thermal effect

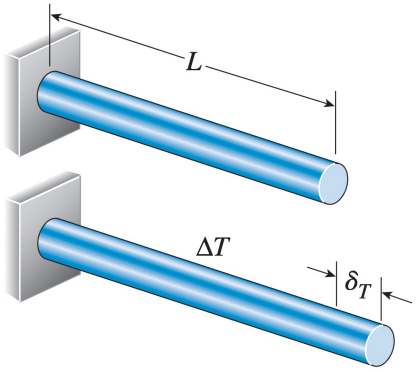
Thermal effect happen due to temperature effect and sunshine. It includes global thermal effect (temperature) and local thermal effect (e.g., due to sunshine).

Thermal strain ε_T will happen when temperature changes for most structural material. α is a property of the material called the *coefficient of thermal expansion*. For statically determinate structure, thermal strain will result in movement of members, but no stress. When the thermal strain is limited in **statically indeterminate structures**, thermal stress will happen.

Temperature-displacement relation:

$$\varepsilon_T = \alpha(\Delta T)$$

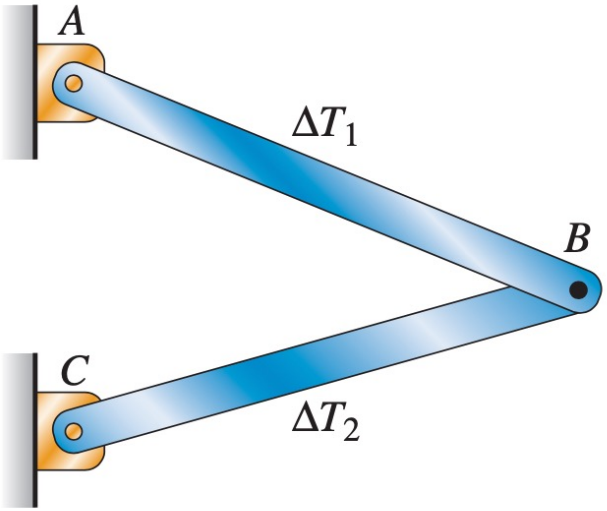
$$\Delta L_T = \alpha(\Delta T)L$$

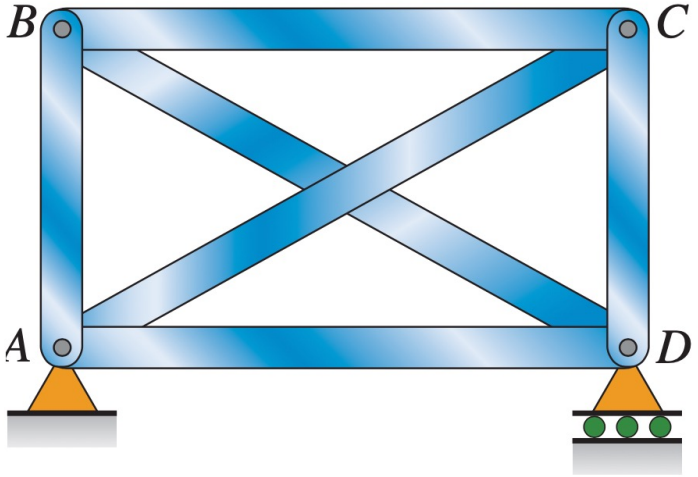


Suppose a beam is fixed at both end and heated, total strain ε is 0, thermal strain ε_T is positive, and compressive mechanical strain will happen, thereby inducing mechanical stress.

$$\varepsilon_{mechanical} = -\alpha(\Delta T) \qquad \sigma = -E \cdot \alpha \Delta T$$

Thermal effect





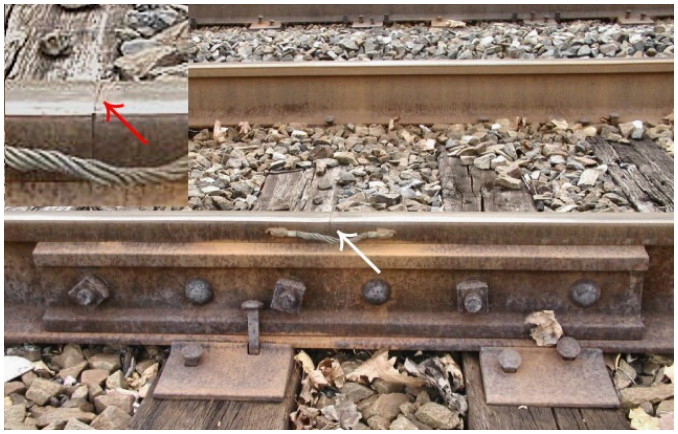
Statically determinant structure: AB increase by ΔT_1 and BC increase by ΔT_2, both bars will deform freely and point B will deform freely. There will be no constraint force due to temperature change. Even when each bar have different temperature. **This apply to almost all statically determinant truss and frames** (considering only axial force and axial displacement).

Statically indeterminant structure: It **may or may not** develop temperature stress, depending on the structure and temperature effect. The top structure has 1 redundant constraint, if all bars are heated uniformly, it just expand and point D will move to right, B move up, no constraint force happen.

Thermal effect

For instance, consider a stainless steel bar with E=210 GPa and $\alpha=1.7\times10^{-5}/^{\circ}$ C. A quick calculation for σ shows that a change in temperature of 60 degree Celsius produces the same strain as a compressive stress of 214 MPa. This stress is in the range of typical allowable stresses for stainless steel. Therefore, **statically indeterminate structures** need to consider thermal effect and designed properly to mitigate thermal effect. Heating may induce compression failure or buckling of member if not properly designed:

$$\sigma = -E \cdot \alpha \Delta T$$

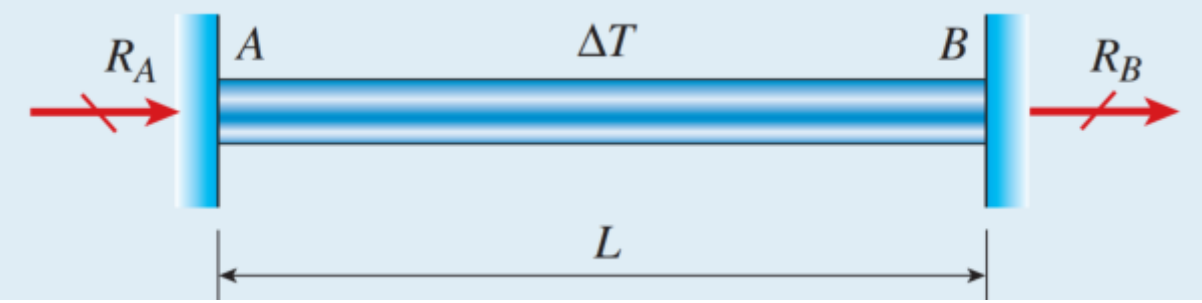




Mitigate thermal stress by human-made gaps

Thermal effect (Example 6)

Consider a uniform bar heated uniformly by temperature and fixed at both ends, the elastic modulus is E, area is A and thermal expansion coefficient is α. (Note: this page is just reformulated of page No. 39)



1. Equation of Equilibrium:

$$R_A + R_B = 0$$

2. Equation of Compatibility:

$$\delta_T + \delta_{mechanical} = 0$$

3. Temperature-displacement relation:

$$\delta_T = \alpha(\Delta T)L$$

4. Force-displacement relationship:

$$\delta_{mechanical} = -\alpha(\Delta T)L = R_B \frac{L}{EA}$$

$$R_A = EA\alpha(\Delta T) \qquad R_B = -EA\alpha(\Delta T)$$

Summary

1. **Displacement, force and stress in axially loaded members**
2. **Displacement of non-uniform cross-section**
3. **Axial stress-strain relationship of typical materials**
4. **Statically indeterminant axial members**
5. **Temperature effect**

THE END